11. Write a program for the task of Credit Score Classification

Aim

To implement a machine learning model for **Credit Score Classification**.

Software Required

- Python 3.x
- Pandas
- Scikit-learn

Procedure

1. Create a credit score dataset.
2. Divide the data into input and target variables.
3. Split the dataset into training and testing sets.
4. Train a classification algorithm.
5. Predict the credit score category.
6. Calculate the accuracy.

Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
data = {
    'Income': [25000, 50000, 75000, 30000, 90000,
               40000, 80000, 20000, 60000, 100000],
    'Debt': [20000, 10000, 5000, 25000, 3000,
             15000, 4000, 30000, 8000, 2000],
    'Age': [22, 30, 40, 25, 45, 35, 42, 21, 38, 50],
    'CreditScore': ['Poor', 'Good', 'Excellent', 'Poor',
                    'Excellent', 'Good', 'Excellent',
                    'Poor', 'Good', 'Excellent']
}
```

```
df = pd.DataFrame(data)
X = df[['Income', 'Debt', 'Age']]
y = df['CreditScore']

X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.3, random_state=42
)
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Predicted Credit Scores:")
print(y_pred)
print("\nAccuracy:",
     accuracy_score(y_test, y_pred))
```

Output:

```
...   Predicted Credit Scores:
      ['Excellent' 'Excellent' 'Poor']

      Accuracy: 0.0
```

Result

Thus, a **Credit Score Classification model** was successfully implemented.